

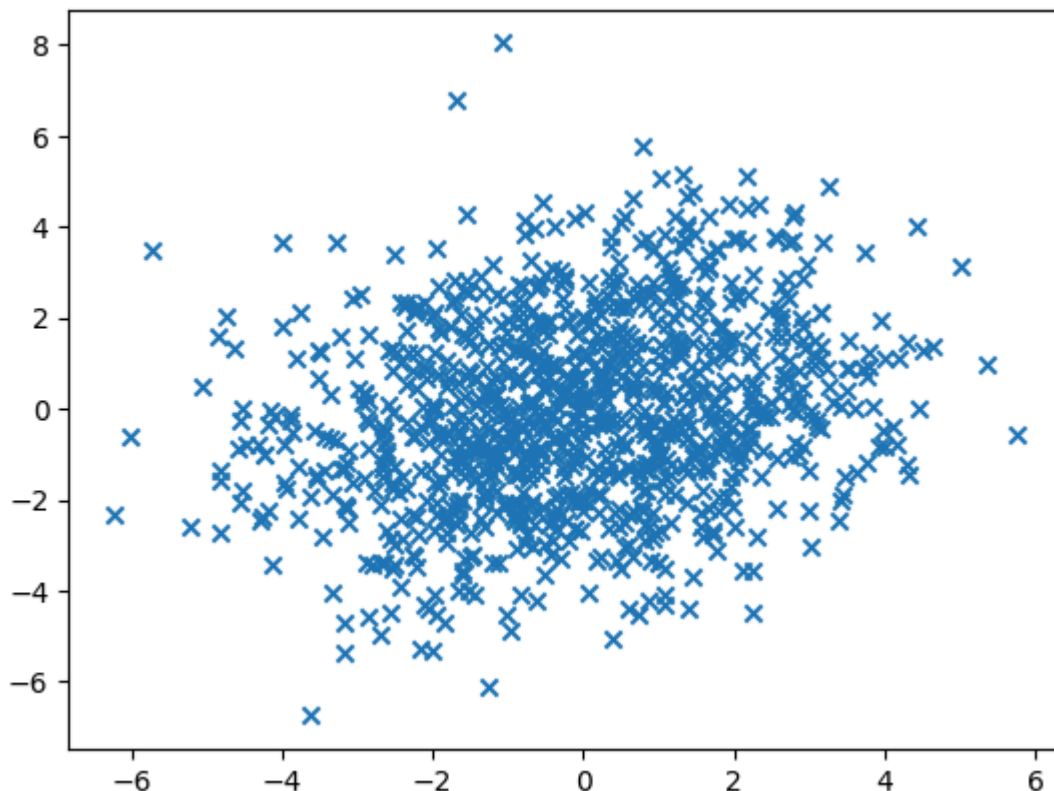
TME06 Bagging, Boosting

Imports

```
In [2]: import numpy as np
from mltools import plot_data, plot_frontiere, make_grid, gen_arti
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.tree import DecisionTreeClassifier
import matplotlib.pyplot as plt
```

Forêts aléatoires (Bagging)

```
In [3]: # Nos données artificielles (2D)
np.random.seed(0)
data, y = gen_arti(centerx=1, centery=1, sigma=0.9, nbex=1000, data_type=0, epsi
plot_data(data)
```



```
In [4]: # Séparation en train/test (70% train, 30% test)
data_train, data_test, y_train, y_test = train_test_split(data, y, test_size=0.3
```

Nous allons tester différents paramètres (`n_estimators` : nombre d'arbres, `max_depth` : profondeur des arbres, `min_samples_split` : nombre minimum d'échantillons pour diviser un nœud, `min_samples_leaf` : nombre minimum d'échantillons par feuille) et étudier l'évolution des frontières de décision, de l'erreur en apprentissage et de l'erreur en test.

```

In [5]: n_estimators_list = [1, 5, 50]
max_depth_list = [None, 1, 5, 50, 100]
min_samples_split_list = [2, 3, 5, 10]
min_samples_leaf_list = [1, 3, 5, 10]

for min_samples_split in min_samples_split_list :
    for min_samples_leaf in min_samples_leaf_list :
        for max_depth in max_depth_list :
            for n_estimators in n_estimators_list :
                classifier = RandomForestClassifier(n_estimators=n_estimators,
                                                  max_depth=max_depth,
                                                  min_samples_split=min_samples_split,
                                                  min_samples_leaf=min_samples_leaf,
                                                  random_state=42)
                classifier.fit(data_train, y_train.ravel()) # On utilise .ravel() pour transformer un array de type ndarray en un array de type numpy.ndarray
                y_pred_train = classifier.predict(data_train)
                y_pred_test = classifier.predict(data_test)

                train_acc = (y_pred_train == y_train.ravel()).mean() * 100
                test_acc = (y_pred_test == y_test.ravel()).mean() * 100

                print(f"Criterion: {classifier.criterion}, Min Split: {min_samples_split}, Max Depth: {max_depth}, N Estimators: {n_estimators}, Train Acc: {train_acc:.2f}, Test Acc: {test_acc:.2f}")

                plot_frontiere(data, lambda data: classifier.predict(data))

```

Criterion: gini, Min Split: 2, Min Leaf: 1, Max Depth: None, N Estimators: 1, Train Acc: 89.29, Test Acc: 68.67
Criterion: gini, Min Split: 2, Min Leaf: 1, Max Depth: None, N Estimators: 5, Train Acc: 97.00, Test Acc: 76.00
Criterion: gini, Min Split: 2, Min Leaf: 1, Max Depth: None, N Estimators: 50, Train Acc: 100.00, Test Acc: 77.67
Criterion: gini, Min Split: 2, Min Leaf: 1, Max Depth: 1, N Estimators: 1, Train Acc: 68.57, Test Acc: 70.00
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Criterion: gini, Min Split: 3, Min Leaf: 5, Max Depth: None, N Estimators: 5, Train Acc: 84.14, Test Acc: 78.67

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Criterion: gini, Min Split: 3, Min Leaf: 10, Max Depth: None, N Estimators: 1, Train Acc: 77.86, Test Acc: 78.33

Criterion: gini, Min Split: 3, Min Leaf: 10, Max Depth: None, N Estimators: 5, Train Acc: 81.00, Test Acc: 81.33

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Criterion: gini, Min Split: 5, Min Leaf: 3, Max Depth: 100, N Estimators: 5, Train Acc: 88.71, Test Acc: 78.00
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Criterion: gini, Min Split: 5, Min Leaf: 5, Max Depth: 1, N Estimators: 5, Train Acc: 78.00, Test Acc: 79.00

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Criterion: gini, Min Split: 5, Min Leaf: 5, Max Depth: 5, N Estimators: 50, Train Acc: 81.29, Test Acc: 80.00

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Criterion: gini, Min Split: 5, Min Leaf: 5, Max Depth: 100, N Estimators: 1, Train Acc: 81.14, Test Acc: 74.33

Criterion: gini, Min Split: 5, Min Leaf: 5, Max Depth: 100, N Estimators: 5, Train Acc: 84.14, Test Acc: 78.67

Criterion: gini, Min Split: 5, Min Leaf: 5, Max Depth: 100, N Estimators: 50, Train Acc: 85.29, Test Acc: 80.33

Criterion: gini, Min Split: 5, Min Leaf: 10, Max Depth: None, N Estimators: 1, Train Acc: 77.86, Test Acc: 78.33

Criterion: gini, Min Split: 5, Min Leaf: 10, Max Depth: None, N Estimators: 5, Train Acc: 81.00, Test Acc: 81.33

Criterion: gini, Min Split: 5, Min Leaf: 10, Max Depth: None, N Estimators: 50, Train Acc: 82.14, Test Acc: 79.67

Criterion: gini, Min Split: 5, Min Leaf: 10, Max Depth: 1, N Estimators: 1, Train Acc: 68.57, Test Acc: 70.00

Criterion: gini, Min Split: 5, Min Leaf: 10, Max Depth: 1, N Estimators: 5, Train Acc: 78.00, Test Acc: 79.00

Criterion: gini, Min Split: 5, Min Leaf: 10, Max Depth: 1, N Estimators: 50, Train Acc: 77.29, Test Acc: 75.67

Criterion: gini, Min Split: 5, Min Leaf: 10, Max Depth: 5, N Estimators: 1, Train Acc: 78.29, Test Acc: 78.67

Criterion: gini, Min Split: 5, Min Leaf: 10, Max Depth: 5, N Estimators: 5, Train Acc: 79.86, Test Acc: 81.00

Criterion: gini, Min Split: 5, Min Leaf: 10, Max Depth: 5, N Estimators: 50, Train Acc: 80.86, Test Acc: 80.33

Criterion: gini, Min Split: 5, Min Leaf: 10, Max Depth: 50, N Estimators: 1, Train Acc: 77.86, Test Acc: 78.33

Criterion: gini, Min Split: 5, Min Leaf: 10, Max Depth: 50, N Estimators: 5, Train Acc: 81.00, Test Acc: 81.33

Criterion: gini, Min Split: 5, Min Leaf: 10, Max Depth: 50, N Estimators: 50, Train Acc: 82.14, Test Acc: 79.67

Criterion: gini, Min Split: 5, Min Leaf: 10, Max Depth: 100, N Estimators: 1, Train Acc: 77.86, Test Acc: 78.33

Criterion: gini, Min Split: 5, Min Leaf: 10, Max Depth: 100, N Estimators: 5, Train Acc: 81.00, Test Acc: 81.33

Criterion: gini, Min Split: 5, Min Leaf: 10, Max Depth: 100, N Estimators: 50, Train Acc: 82.14, Test Acc: 79.67

Criterion: gini, Min Split: 10, Min Leaf: 1, Max Depth: None, N Estimators: 1, Train Acc: 81.43, Test Acc: 73.67
Criterion: gini, Min Split: 10, Min Leaf: 1, Max Depth: None, N Estimators: 5, Train Acc: 89.86, Test Acc: 77.67
Criterion: gini, Min Split: 10, Min Leaf: 1, Max Depth: None, N Estimators: 50, Train Acc: 90.43, Test Acc: 79.00
Criterion: gini, Min Split: 10, Min Leaf: 1, Max Depth: 1, N Estimators: 1, Train Acc: 68.57, Test Acc: 70.00
Criterion: gini, Min Split: 10, Min Leaf: 1, Max Depth: 1, N Estimators: 5, Train Acc: 78.00, Test Acc: 79.00
Criterion: gini, Min Split: 10, Min Leaf: 1, Max Depth: 1, N Estimators: 50, Train Acc: 77.29, Test Acc: 75.67
Criterion: gini, Min Split: 10, Min Leaf: 1, Max Depth: 5, N Estimators: 1, Train Acc: 78.71, Test Acc: 76.67
Criterion: gini, Min Split: 10, Min Leaf: 1, Max Depth: 5, N Estimators: 5, Train Acc: 82.29, Test Acc: 80.67
Criterion: gini, Min Split: 10, Min Leaf: 1, Max Depth: 5, N Estimators: 50, Train Acc: 81.29, Test Acc: 79.67
Criterion: gini, Min Split: 10, Min Leaf: 1, Max Depth: 50, N Estimators: 1, Train Acc: 81.43, Test Acc: 73.67
Criterion: gini, Min Split: 10, Min Leaf: 1, Max Depth: 50, N Estimators: 5, Train Acc: 89.86, Test Acc: 77.67
Criterion: gini, Min Split: 10, Min Leaf: 1, Max Depth: 50, N Estimators: 50, Train Acc: 90.43, Test Acc: 79.00
Criterion: gini, Min Split: 10, Min Leaf: 1, Max Depth: 100, N Estimators: 1, Train Acc: 81.43, Test Acc: 73.67
Criterion: gini, Min Split: 10, Min Leaf: 1, Max Depth: 100, N Estimators: 5, Train Acc: 89.86, Test Acc: 77.67
Criterion: gini, Min Split: 10, Min Leaf: 1, Max Depth: 100, N Estimators: 50, Train Acc: 90.43, Test Acc: 79.00
Criterion: gini, Min Split: 10, Min Leaf: 3, Max Depth: None, N Estimators: 1, Train Acc: 79.71, Test Acc: 75.33
Criterion: gini, Min Split: 10, Min Leaf: 3, Max Depth: None, N Estimators: 5, Train Acc: 85.86, Test Acc: 77.00
Criterion: gini, Min Split: 10, Min Leaf: 3, Max Depth: None, N Estimators: 50, Train Acc: 88.14, Test Acc: 80.67
Criterion: gini, Min Split: 10, Min Leaf: 3, Max Depth: 1, N Estimators: 1, Train Acc: 68.57, Test Acc: 70.00
Criterion: gini, Min Split: 10, Min Leaf: 3, Max Depth: 1, N Estimators: 5, Train Acc: 78.00, Test Acc: 79.00
Criterion: gini, Min Split: 10, Min Leaf: 3, Max Depth: 1, N Estimators: 50, Train Acc: 77.29, Test Acc: 75.67
Criterion: gini, Min Split: 10, Min Leaf: 3, Max Depth: 5, N Estimators: 1, Train Acc: 78.43, Test Acc: 76.33
Criterion: gini, Min Split: 10, Min Leaf: 3, Max Depth: 5, N Estimators: 5, Train Acc: 80.71, Test Acc: 81.00
Criterion: gini, Min Split: 10, Min Leaf: 3, Max Depth: 5, N Estimators: 50, Train Acc: 81.57, Test Acc: 79.67
Criterion: gini, Min Split: 10, Min Leaf: 3, Max Depth: 50, N Estimators: 1, Train Acc: 79.71, Test Acc: 75.33
Criterion: gini, Min Split: 10, Min Leaf: 3, Max Depth: 50, N Estimators: 5, Train Acc: 85.86, Test Acc: 77.00
Criterion: gini, Min Split: 10, Min Leaf: 3, Max Depth: 50, N Estimators: 50, Train Acc: 88.14, Test Acc: 80.67
Criterion: gini, Min Split: 10, Min Leaf: 3, Max Depth: 100, N Estimators: 1, Train Acc: 79.71, Test Acc: 75.33
Criterion: gini, Min Split: 10, Min Leaf: 3, Max Depth: 100, N Estimators: 5, Train Acc: 85.86, Test Acc: 77.00
Criterion: gini, Min Split: 10, Min Leaf: 3, Max Depth: 100, N Estimators: 50, Train Acc: 88.14, Test Acc: 80.67

Criterion: gini, Min Split: 10, Min Leaf: 5, Max Depth: None, N Estimators: 1, Train Acc: 81.14, Test Acc: 74.33

Criterion: gini, Min Split: 10, Min Leaf: 5, Max Depth: None, N Estimators: 5, Train Acc: 84.14, Test Acc: 78.67

Criterion: gini, Min Split: 10, Min Leaf: 5, Max Depth: None, N Estimators: 50, Train Acc: 85.29, Test Acc: 80.33

Criterion: gini, Min Split: 10, Min Leaf: 5, Max Depth: 1, N Estimators: 1, Train Acc: 68.57, Test Acc: 70.00

Criterion: gini, Min Split: 10, Min Leaf: 5, Max Depth: 1, N Estimators: 5, Train Acc: 78.00, Test Acc: 79.00

Criterion: gini, Min Split: 10, Min Leaf: 5, Max Depth: 1, N Estimators: 50, Train Acc: 77.29, Test Acc: 75.67

Criterion: gini, Min Split: 10, Min Leaf: 5, Max Depth: 5, N Estimators: 1, Train Acc: 79.00, Test Acc: 79.33

Criterion: gini, Min Split: 10, Min Leaf: 5, Max Depth: 5, N Estimators: 5, Train Acc: 80.57, Test Acc: 82.00

Criterion: gini, Min Split: 10, Min Leaf: 5, Max Depth: 5, N Estimators: 50, Train Acc: 81.29, Test Acc: 80.00

Criterion: gini, Min Split: 10, Min Leaf: 5, Max Depth: 50, N Estimators: 1, Train Acc: 81.14, Test Acc: 74.33

Criterion: gini, Min Split: 10, Min Leaf: 5, Max Depth: 50, N Estimators: 5, Train Acc: 84.14, Test Acc: 78.67

Criterion: gini, Min Split: 10, Min Leaf: 5, Max Depth: 50, N Estimators: 50, Train Acc: 85.29, Test Acc: 80.33

Criterion: gini, Min Split: 10, Min Leaf: 5, Max Depth: 100, N Estimators: 1, Train Acc: 81.14, Test Acc: 74.33

Criterion: gini, Min Split: 10, Min Leaf: 5, Max Depth: 100, N Estimators: 5, Train Acc: 84.14, Test Acc: 78.67

Criterion: gini, Min Split: 10, Min Leaf: 5, Max Depth: 100, N Estimators: 50, Train Acc: 85.29, Test Acc: 80.33

Criterion: gini, Min Split: 10, Min Leaf: 10, Max Depth: None, N Estimators: 1, Train Acc: 77.86, Test Acc: 78.33

Criterion: gini, Min Split: 10, Min Leaf: 10, Max Depth: None, N Estimators: 5, Train Acc: 81.00, Test Acc: 81.33

Criterion: gini, Min Split: 10, Min Leaf: 10, Max Depth: None, N Estimators: 50, Train Acc: 82.14, Test Acc: 79.67

Criterion: gini, Min Split: 10, Min Leaf: 10, Max Depth: 1, N Estimators: 1, Train Acc: 68.57, Test Acc: 70.00

Criterion: gini, Min Split: 10, Min Leaf: 10, Max Depth: 1, N Estimators: 5, Train Acc: 78.00, Test Acc: 79.00

Criterion: gini, Min Split: 10, Min Leaf: 10, Max Depth: 1, N Estimators: 50, Train Acc: 77.29, Test Acc: 75.67

Criterion: gini, Min Split: 10, Min Leaf: 10, Max Depth: 5, N Estimators: 1, Train Acc: 78.29, Test Acc: 78.67

Criterion: gini, Min Split: 10, Min Leaf: 10, Max Depth: 5, N Estimators: 5, Train Acc: 79.86, Test Acc: 81.00

Criterion: gini, Min Split: 10, Min Leaf: 10, Max Depth: 5, N Estimators: 50, Train Acc: 80.86, Test Acc: 80.33

Criterion: gini, Min Split: 10, Min Leaf: 10, Max Depth: 50, N Estimators: 1, Train Acc: 77.86, Test Acc: 78.33

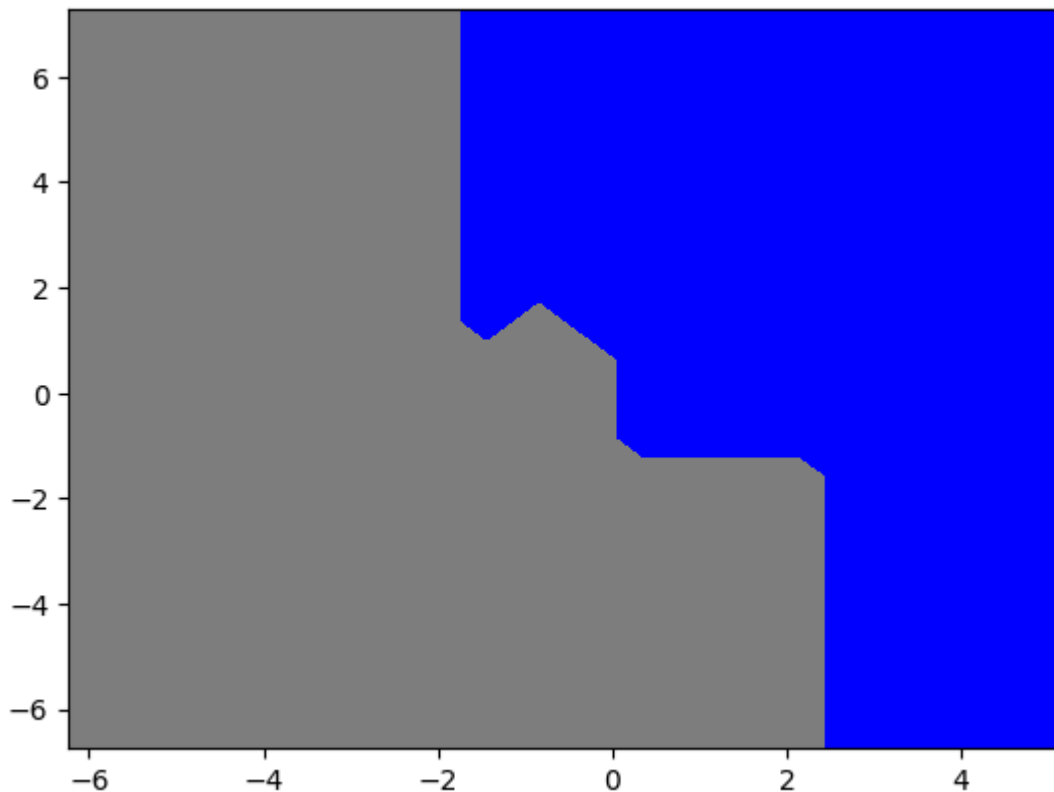
Criterion: gini, Min Split: 10, Min Leaf: 10, Max Depth: 50, N Estimators: 5, Train Acc: 81.00, Test Acc: 81.33

Criterion: gini, Min Split: 10, Min Leaf: 10, Max Depth: 50, N Estimators: 50, Train Acc: 82.14, Test Acc: 79.67

Criterion: gini, Min Split: 10, Min Leaf: 10, Max Depth: 100, N Estimators: 1, Train Acc: 77.86, Test Acc: 78.33

Criterion: gini, Min Split: 10, Min Leaf: 10, Max Depth: 100, N Estimators: 5, Train Acc: 81.00, Test Acc: 81.33

Criterion: gini, Min Split: 10, Min Leaf: 10, Max Depth: 100, N Estimators: 50, Train Acc: 82.14, Test Acc: 79.67



On observe que sur des données bruitées, plus la profondeur des arbres et le nombre d'arbres des forêts aléatoires augmentent, plus il y a du surapprentissage. Par exemple, pour un modèle avec un nombre minimum d'échantillons par feuille égal à 1, une profondeur de 100 et 50 arbres, l'accuracy sur le train atteint 100%, tandis que sur le test, elle tombe à 77.67%. Cela montre un écart important entre les données d'entraînement et les données de test, typique du surapprentissage : le modèle est trop ajusté aux données d'entraînement et n'arrive pas à généraliser correctement aux nouvelles données.

Cependant, ce problème de surapprentissage peut être atténué en augmentant le nombre minimum d'échantillons par feuille ou le nombre minimum d'échantillons requis pour diviser un nœud. En faisant cela, on voit que l'accuracy devient plus stable/similaire entre les données d'entraînement et de test. Cela signifie que le modèle ne surapprend plus et devient plus généraliste. On note cependant que l'accuracy (sur train et test) se situe dans une plage de 70-80%. Cela montre que le modèle Random Forest est relativement robuste au surapprentissage grâce au bagging (sous-échantillonnage). En effet, le modèle combine les résultats de plusieurs arbres, ce qui permet de lisser les erreurs : même si un arbre surapprend, la moyenne des prédictions des différents arbres permet de réduire cet effet de surapprentissage.

AdaBoost (Boosting)

Nous allons maintenant passer à l'implémentation d'AdaBoost et observer l'évolution de l'erreur ϵ_t , des poids α_t et de Z_t , le cumul des fonctions de normalisation Z_t .

```
In [6]: def AdaBoost_algorithm(X, Y, T=50, max_depth=1):
        N, _ = X.shape
```

```

D = np.ones(N) / N
classifiers = []
alphas = []
epsilons = []
Z_values = []

for t in range(T):
    # Apprentissage d'un arbre faible
    stump = DecisionTreeClassifier(max_depth=max_depth)
    stump.fit(X, Y, sample_weight=D)
    pred = stump.predict(X)

    # Calcul de l'erreur epsilon_t
    Y = Y.flatten()
    miss = (pred != Y)
    epsilon_t = max(np.dot(D, miss), 1e-10)

    # Fixation de alpha_t
    alpha_t = 0.5 * np.log((1 - epsilon_t) / epsilon_t)

    # Mise à jour des poids
    D = D * np.exp(-alpha_t * Y * pred)
    Z_t = np.sum(D)
    D /= Z_t

    classifiers.append(stump)
    alphas.append(alpha_t)
    epsilons.append(epsilon_t)
    Z_values.append(Z_t)

    print(f"Iteration {t+1}: epsilon_t = {epsilon_t:.4f}, alpha_t = {alpha_t:.4f}")

return classifiers, alphas, epsilons, Z_values

```

```

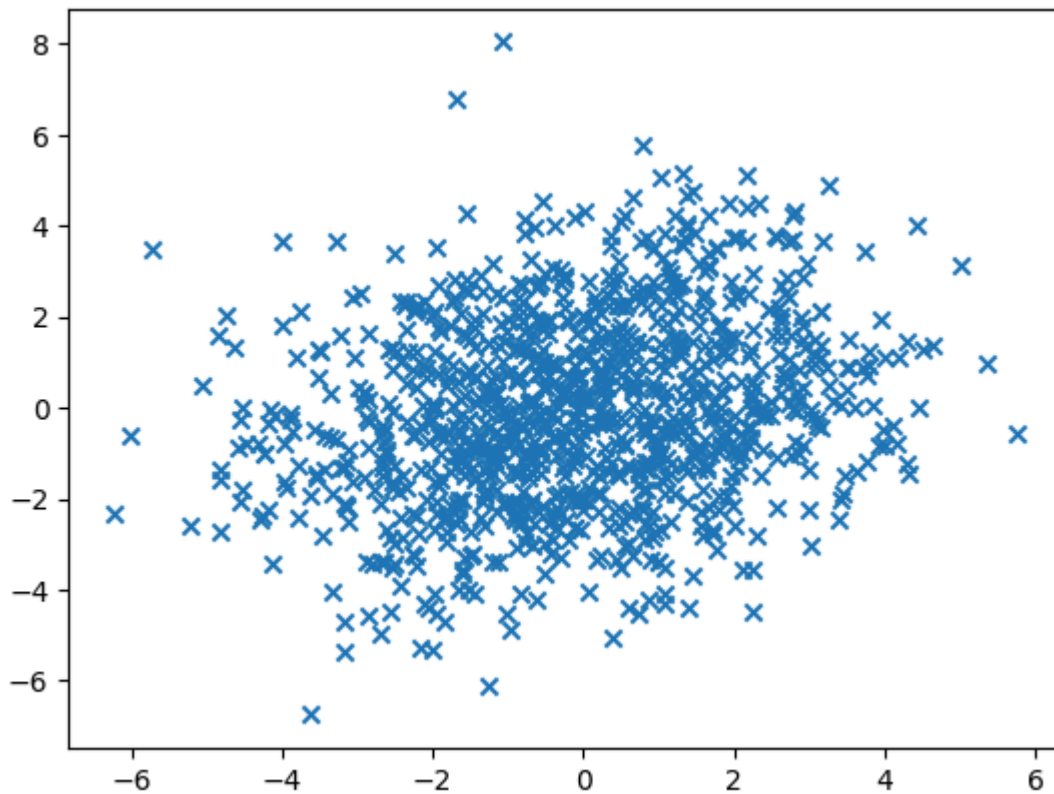
In [7]: def AdaBoost_classifier(X, classifiers, alphas):
        clf_preds = np.array([alpha * classifier.predict(X) for classifier, alpha in
                               zip(classifiers, alphas)])
        return np.sign(np.sum(clf_preds, axis=0))

```

```

In [15]: # Nos données artificielles (2D)
np.random.seed(0)
data, y = gen_arti(centerx=1, centery=1, sigma=0.9, nbex=1000, data_type=0, epsilon=0.1)
plot_data(data)

```



```
In [16]: # Séparation en train/test (70% train, 30% test)
data_train, data_test, y_train, y_test = train_test_split(data, y, test_size=0.3
```

```
In [17]: # MAX_DEPTH = 1

# Entraînement sur train
classifiers, alphas, epsilons, Z_values = AdaBoost_algorithm(data_train, y_train

# Prédiction sur train
y_pred_train = AdaBoost_classifient(data_train, classifiers, alphas)
train_acc = (y_pred_train == y_train.ravel()).mean() * 100
print("Accuracy sur les données train:", train_acc)

# Prédiction sur test
y_pred_test = AdaBoost_classifient(data_test, classifiers, alphas)
test_acc = (y_pred_test == y_test.ravel()).mean() * 100
print("Accuracy sur les données test:", test_acc)
```

Iteration 1: epsilon_t = 0.2629, alpha_t = 0.5156, Z_t = 0.8804
Iteration 2: epsilon_t = 0.3101, alpha_t = 0.3997, Z_t = 0.9251
Iteration 3: epsilon_t = 0.4384, alpha_t = 0.1238, Z_t = 0.9924
Iteration 4: epsilon_t = 0.3891, alpha_t = 0.2255, Z_t = 0.9751
Iteration 5: epsilon_t = 0.4657, alpha_t = 0.0687, Z_t = 0.9976
Iteration 6: epsilon_t = 0.4538, alpha_t = 0.0926, Z_t = 0.9957
Iteration 7: epsilon_t = 0.4235, alpha_t = 0.1543, Z_t = 0.9882
Iteration 8: epsilon_t = 0.4262, alpha_t = 0.1487, Z_t = 0.9891
Iteration 9: epsilon_t = 0.4282, alpha_t = 0.1446, Z_t = 0.9896
Iteration 10: epsilon_t = 0.4372, alpha_t = 0.1263, Z_t = 0.9921
Iteration 11: epsilon_t = 0.4406, alpha_t = 0.1194, Z_t = 0.9929
Iteration 12: epsilon_t = 0.4391, alpha_t = 0.1225, Z_t = 0.9925
Iteration 13: epsilon_t = 0.4747, alpha_t = 0.0507, Z_t = 0.9987
Iteration 14: epsilon_t = 0.4860, alpha_t = 0.0279, Z_t = 0.9996
Iteration 15: epsilon_t = 0.4864, alpha_t = 0.0272, Z_t = 0.9996
Iteration 16: epsilon_t = 0.4747, alpha_t = 0.0507, Z_t = 0.9987
Iteration 17: epsilon_t = 0.4759, alpha_t = 0.0482, Z_t = 0.9988
Iteration 18: epsilon_t = 0.4432, alpha_t = 0.1140, Z_t = 0.9935
Iteration 19: epsilon_t = 0.4803, alpha_t = 0.0395, Z_t = 0.9992
Iteration 20: epsilon_t = 0.4776, alpha_t = 0.0449, Z_t = 0.9990
Iteration 21: epsilon_t = 0.4778, alpha_t = 0.0444, Z_t = 0.9990
Iteration 22: epsilon_t = 0.4788, alpha_t = 0.0425, Z_t = 0.9991
Iteration 23: epsilon_t = 0.4794, alpha_t = 0.0413, Z_t = 0.9991
Iteration 24: epsilon_t = 0.4898, alpha_t = 0.0205, Z_t = 0.9998
Iteration 25: epsilon_t = 0.4894, alpha_t = 0.0212, Z_t = 0.9998
Iteration 26: epsilon_t = 0.4800, alpha_t = 0.0400, Z_t = 0.9992
Iteration 27: epsilon_t = 0.4808, alpha_t = 0.0384, Z_t = 0.9993
Iteration 28: epsilon_t = 0.4883, alpha_t = 0.0233, Z_t = 0.9997
Iteration 29: epsilon_t = 0.4476, alpha_t = 0.1051, Z_t = 0.9945
Iteration 30: epsilon_t = 0.4599, alpha_t = 0.0804, Z_t = 0.9968
Iteration 31: epsilon_t = 0.4684, alpha_t = 0.0633, Z_t = 0.9980
Iteration 32: epsilon_t = 0.4828, alpha_t = 0.0344, Z_t = 0.9994
Iteration 33: epsilon_t = 0.4913, alpha_t = 0.0174, Z_t = 0.9998
Iteration 34: epsilon_t = 0.4915, alpha_t = 0.0171, Z_t = 0.9999
Iteration 35: epsilon_t = 0.4916, alpha_t = 0.0168, Z_t = 0.9999
Iteration 36: epsilon_t = 0.4916, alpha_t = 0.0168, Z_t = 0.9999
Iteration 37: epsilon_t = 0.4838, alpha_t = 0.0325, Z_t = 0.9995
Iteration 38: epsilon_t = 0.4842, alpha_t = 0.0317, Z_t = 0.9995
Iteration 39: epsilon_t = 0.4921, alpha_t = 0.0158, Z_t = 0.9999
Iteration 40: epsilon_t = 0.4919, alpha_t = 0.0163, Z_t = 0.9999
Iteration 41: epsilon_t = 0.4920, alpha_t = 0.0160, Z_t = 0.9999
Iteration 42: epsilon_t = 0.4921, alpha_t = 0.0158, Z_t = 0.9999
Iteration 43: epsilon_t = 0.4848, alpha_t = 0.0305, Z_t = 0.9995
Iteration 44: epsilon_t = 0.4851, alpha_t = 0.0298, Z_t = 0.9996
Iteration 45: epsilon_t = 0.4926, alpha_t = 0.0149, Z_t = 0.9999
Iteration 46: epsilon_t = 0.4923, alpha_t = 0.0153, Z_t = 0.9999
Iteration 47: epsilon_t = 0.4925, alpha_t = 0.0151, Z_t = 0.9999
Iteration 48: epsilon_t = 0.4926, alpha_t = 0.0149, Z_t = 0.9999
Iteration 49: epsilon_t = 0.4856, alpha_t = 0.0288, Z_t = 0.9996
Iteration 50: epsilon_t = 0.4859, alpha_t = 0.0282, Z_t = 0.9996
Iteration 51: epsilon_t = 0.4859, alpha_t = 0.0282, Z_t = 0.9996
Iteration 52: epsilon_t = 0.4860, alpha_t = 0.0279, Z_t = 0.9996
Iteration 53: epsilon_t = 0.4864, alpha_t = 0.0272, Z_t = 0.9996
Iteration 54: epsilon_t = 0.4866, alpha_t = 0.0267, Z_t = 0.9996
Iteration 55: epsilon_t = 0.4866, alpha_t = 0.0267, Z_t = 0.9996
Iteration 56: epsilon_t = 0.4868, alpha_t = 0.0265, Z_t = 0.9996
Iteration 57: epsilon_t = 0.4590, alpha_t = 0.0821, Z_t = 0.9966
Iteration 58: epsilon_t = 0.4724, alpha_t = 0.0553, Z_t = 0.9985
Iteration 59: epsilon_t = 0.4772, alpha_t = 0.0457, Z_t = 0.9990
Iteration 60: epsilon_t = 0.4870, alpha_t = 0.0260, Z_t = 0.9997

```

Iteration 61: epsilon_t = 0.4872, alpha_t = 0.0257, Z_t = 0.9997
Iteration 62: epsilon_t = 0.4874, alpha_t = 0.0251, Z_t = 0.9997
Iteration 63: epsilon_t = 0.4876, alpha_t = 0.0249, Z_t = 0.9997
Iteration 64: epsilon_t = 0.4876, alpha_t = 0.0248, Z_t = 0.9997
Iteration 65: epsilon_t = 0.4878, alpha_t = 0.0244, Z_t = 0.9997
Iteration 66: epsilon_t = 0.4880, alpha_t = 0.0239, Z_t = 0.9997
Iteration 67: epsilon_t = 0.4881, alpha_t = 0.0237, Z_t = 0.9997
Iteration 68: epsilon_t = 0.4882, alpha_t = 0.0236, Z_t = 0.9997
Iteration 69: epsilon_t = 0.4884, alpha_t = 0.0233, Z_t = 0.9997
Iteration 70: epsilon_t = 0.4886, alpha_t = 0.0229, Z_t = 0.9997
Iteration 71: epsilon_t = 0.4887, alpha_t = 0.0227, Z_t = 0.9997
Iteration 72: epsilon_t = 0.4887, alpha_t = 0.0226, Z_t = 0.9997
Iteration 73: epsilon_t = 0.4889, alpha_t = 0.0223, Z_t = 0.9998
Iteration 74: epsilon_t = 0.4891, alpha_t = 0.0219, Z_t = 0.9998
Iteration 75: epsilon_t = 0.4892, alpha_t = 0.0217, Z_t = 0.9998
Iteration 76: epsilon_t = 0.4892, alpha_t = 0.0216, Z_t = 0.9998
Iteration 77: epsilon_t = 0.4893, alpha_t = 0.0213, Z_t = 0.9998
Iteration 78: epsilon_t = 0.4895, alpha_t = 0.0210, Z_t = 0.9998
Iteration 79: epsilon_t = 0.4896, alpha_t = 0.0208, Z_t = 0.9998
Iteration 80: epsilon_t = 0.4896, alpha_t = 0.0207, Z_t = 0.9998
Iteration 81: epsilon_t = 0.4898, alpha_t = 0.0205, Z_t = 0.9998
Iteration 82: epsilon_t = 0.4899, alpha_t = 0.0201, Z_t = 0.9998
Iteration 83: epsilon_t = 0.4900, alpha_t = 0.0200, Z_t = 0.9998
Iteration 84: epsilon_t = 0.4900, alpha_t = 0.0199, Z_t = 0.9998
Iteration 85: epsilon_t = 0.4902, alpha_t = 0.0197, Z_t = 0.9998
Iteration 86: epsilon_t = 0.4780, alpha_t = 0.0440, Z_t = 0.9990
Iteration 87: epsilon_t = 0.4751, alpha_t = 0.0497, Z_t = 0.9988
Iteration 88: epsilon_t = 0.4909, alpha_t = 0.0182, Z_t = 0.9998
Iteration 89: epsilon_t = 0.4906, alpha_t = 0.0187, Z_t = 0.9998
Iteration 90: epsilon_t = 0.4908, alpha_t = 0.0184, Z_t = 0.9998
Iteration 91: epsilon_t = 0.4910, alpha_t = 0.0181, Z_t = 0.9998
Iteration 92: epsilon_t = 0.4910, alpha_t = 0.0181, Z_t = 0.9998
Iteration 93: epsilon_t = 0.4911, alpha_t = 0.0178, Z_t = 0.9998
Iteration 94: epsilon_t = 0.4913, alpha_t = 0.0174, Z_t = 0.9998
Iteration 95: epsilon_t = 0.4914, alpha_t = 0.0173, Z_t = 0.9999
Iteration 96: epsilon_t = 0.4914, alpha_t = 0.0173, Z_t = 0.9999
Iteration 97: epsilon_t = 0.4914, alpha_t = 0.0172, Z_t = 0.9999
Iteration 98: epsilon_t = 0.4916, alpha_t = 0.0169, Z_t = 0.9999
Iteration 99: epsilon_t = 0.4916, alpha_t = 0.0167, Z_t = 0.9999
Iteration 100: epsilon_t = 0.4916, alpha_t = 0.0167, Z_t = 0.9999
Accuracy sur les données train: 78.0
Accuracy sur les données test: 79.0

```

```

In [18]: Z = np.cumprod(Z_values)

plt.figure(figsize=(12, 8))

plt.subplot(2, 2, 1)
plt.plot(epsilons)
plt.title("Erreur  $\epsilon_t$  par itération")
plt.xlabel("Itération t")
plt.ylabel(" $\epsilon_t$ ")

plt.subplot(2, 2, 2)
plt.plot(alphas)
plt.title("Poids  $\alpha_t$  par itération")
plt.xlabel("Itération t")
plt.ylabel(" $\alpha_t$ ")

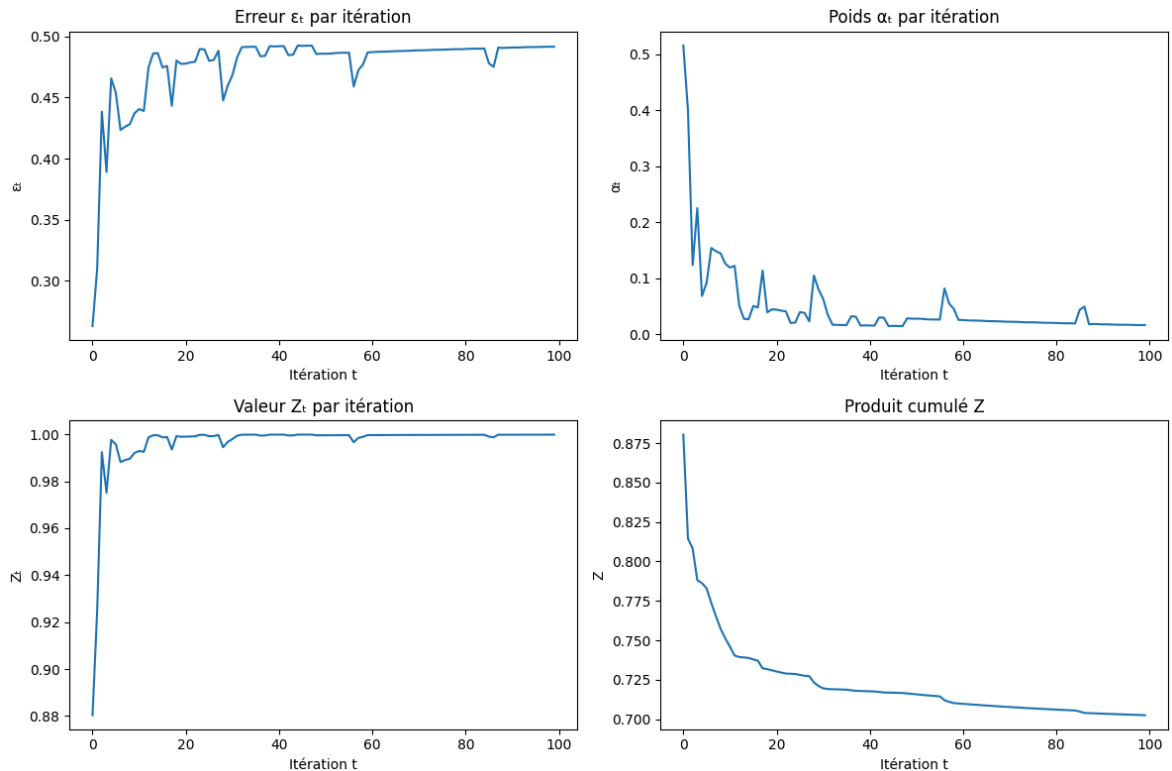
plt.subplot(2, 2, 3)

```

```
plt.plot(Z_values)
plt.title("Valeur  $Z_t$  par itération")
plt.xlabel("Itération  $t$ ")
plt.ylabel(" $Z_t$ ")

plt.subplot(2, 2, 4)
plt.plot(Z)
plt.title("Produit cumulé  $Z$ ")
plt.xlabel("Itération  $t$ ")
plt.ylabel(" $Z$ ")

plt.tight_layout()
plt.show()
```



```
In [19]: # MAX_DEPTH = 5

# Entraînement sur train
classifiers, alphas, epsilons, Z_values = AdaBoost_algorithm(data_train, y_train)

# Prédiction sur train
y_pred_train = AdaBoost_classifier(data_train, classifiers, alphas)
train_acc = (y_pred_train == y_train.ravel()).mean() * 100
print("Accuracy sur les données train:", train_acc)

# Prédiction sur test
y_pred_test = AdaBoost_classifier(data_test, classifiers, alphas)
test_acc = (y_pred_test == y_test.ravel()).mean() * 100
print("Accuracy sur les données test:", test_acc)
```


Iteration 1: epsilon_t = 0.1814, alpha_t = 0.7533, Z_t = 0.7707
Iteration 2: epsilon_t = 0.2527, alpha_t = 0.5421, Z_t = 0.8691
Iteration 3: epsilon_t = 0.2508, alpha_t = 0.5471, Z_t = 0.8670
Iteration 4: epsilon_t = 0.3464, alpha_t = 0.3174, Z_t = 0.9517
Iteration 5: epsilon_t = 0.3375, alpha_t = 0.3371, Z_t = 0.9457
Iteration 6: epsilon_t = 0.3562, alpha_t = 0.2960, Z_t = 0.9577
Iteration 7: epsilon_t = 0.4032, alpha_t = 0.1961, Z_t = 0.9811
Iteration 8: epsilon_t = 0.4356, alpha_t = 0.1296, Z_t = 0.9917
Iteration 9: epsilon_t = 0.4340, alpha_t = 0.1328, Z_t = 0.9912
Iteration 10: epsilon_t = 0.3934, alpha_t = 0.2166, Z_t = 0.9770
Iteration 11: epsilon_t = 0.4145, alpha_t = 0.1727, Z_t = 0.9853
Iteration 12: epsilon_t = 0.4064, alpha_t = 0.1894, Z_t = 0.9823
Iteration 13: epsilon_t = 0.4017, alpha_t = 0.1991, Z_t = 0.9805
Iteration 14: epsilon_t = 0.3910, alpha_t = 0.2216, Z_t = 0.9759
Iteration 15: epsilon_t = 0.4113, alpha_t = 0.1793, Z_t = 0.9841
Iteration 16: epsilon_t = 0.4345, alpha_t = 0.1318, Z_t = 0.9914
Iteration 17: epsilon_t = 0.3966, alpha_t = 0.2099, Z_t = 0.9784
Iteration 18: epsilon_t = 0.4143, alpha_t = 0.1731, Z_t = 0.9852
Iteration 19: epsilon_t = 0.4017, alpha_t = 0.1992, Z_t = 0.9805
Iteration 20: epsilon_t = 0.3491, alpha_t = 0.3115, Z_t = 0.9534
Iteration 21: epsilon_t = 0.3690, alpha_t = 0.2682, Z_t = 0.9651
Iteration 22: epsilon_t = 0.3542, alpha_t = 0.3003, Z_t = 0.9566
Iteration 23: epsilon_t = 0.3436, alpha_t = 0.3236, Z_t = 0.9498
Iteration 24: epsilon_t = 0.3559, alpha_t = 0.2967, Z_t = 0.9575
Iteration 25: epsilon_t = 0.3793, alpha_t = 0.2463, Z_t = 0.9704
Iteration 26: epsilon_t = 0.3810, alpha_t = 0.2427, Z_t = 0.9712
Iteration 27: epsilon_t = 0.3580, alpha_t = 0.2920, Z_t = 0.9588
Iteration 28: epsilon_t = 0.3412, alpha_t = 0.3290, Z_t = 0.9482
Iteration 29: epsilon_t = 0.3520, alpha_t = 0.3051, Z_t = 0.9552
Iteration 30: epsilon_t = 0.3975, alpha_t = 0.2079, Z_t = 0.9788
Iteration 31: epsilon_t = 0.3548, alpha_t = 0.2989, Z_t = 0.9569
Iteration 32: epsilon_t = 0.3552, alpha_t = 0.2981, Z_t = 0.9572
Iteration 33: epsilon_t = 0.4293, alpha_t = 0.1423, Z_t = 0.9900
Iteration 34: epsilon_t = 0.4012, alpha_t = 0.2002, Z_t = 0.9803
Iteration 35: epsilon_t = 0.4086, alpha_t = 0.1848, Z_t = 0.9832
Iteration 36: epsilon_t = 0.4104, alpha_t = 0.1812, Z_t = 0.9838
Iteration 37: epsilon_t = 0.4226, alpha_t = 0.1560, Z_t = 0.9880
Iteration 38: epsilon_t = 0.4178, alpha_t = 0.1659, Z_t = 0.9864
Iteration 39: epsilon_t = 0.3876, alpha_t = 0.2286, Z_t = 0.9744
Iteration 40: epsilon_t = 0.3969, alpha_t = 0.2092, Z_t = 0.9785
Iteration 41: epsilon_t = 0.3887, alpha_t = 0.2264, Z_t = 0.9749
Iteration 42: epsilon_t = 0.4175, alpha_t = 0.1666, Z_t = 0.9863
Iteration 43: epsilon_t = 0.4122, alpha_t = 0.1775, Z_t = 0.9844
Iteration 44: epsilon_t = 0.4278, alpha_t = 0.1453, Z_t = 0.9895
Iteration 45: epsilon_t = 0.4226, alpha_t = 0.1561, Z_t = 0.9879
Iteration 46: epsilon_t = 0.4085, alpha_t = 0.1850, Z_t = 0.9831
Iteration 47: epsilon_t = 0.4019, alpha_t = 0.1987, Z_t = 0.9806
Iteration 48: epsilon_t = 0.4311, alpha_t = 0.1387, Z_t = 0.9905
Iteration 49: epsilon_t = 0.4302, alpha_t = 0.1405, Z_t = 0.9902
Iteration 50: epsilon_t = 0.4156, alpha_t = 0.1704, Z_t = 0.9857
Iteration 51: epsilon_t = 0.3708, alpha_t = 0.2643, Z_t = 0.9661
Iteration 52: epsilon_t = 0.4052, alpha_t = 0.1919, Z_t = 0.9819
Iteration 53: epsilon_t = 0.3977, alpha_t = 0.2076, Z_t = 0.9788
Iteration 54: epsilon_t = 0.4189, alpha_t = 0.1636, Z_t = 0.9868
Iteration 55: epsilon_t = 0.3965, alpha_t = 0.2101, Z_t = 0.9783
Iteration 56: epsilon_t = 0.4093, alpha_t = 0.1835, Z_t = 0.9834
Iteration 57: epsilon_t = 0.3954, alpha_t = 0.2124, Z_t = 0.9779
Iteration 58: epsilon_t = 0.4179, alpha_t = 0.1658, Z_t = 0.9864
Iteration 59: epsilon_t = 0.4219, alpha_t = 0.1574, Z_t = 0.9877
Iteration 60: epsilon_t = 0.4378, alpha_t = 0.1251, Z_t = 0.9922

```

Iteration 61: epsilon_t = 0.4090, alpha_t = 0.1840, Z_t = 0.9833
Iteration 62: epsilon_t = 0.4055, alpha_t = 0.1913, Z_t = 0.9820
Iteration 63: epsilon_t = 0.4002, alpha_t = 0.2023, Z_t = 0.9799
Iteration 64: epsilon_t = 0.4077, alpha_t = 0.1868, Z_t = 0.9828
Iteration 65: epsilon_t = 0.4107, alpha_t = 0.1804, Z_t = 0.9839
Iteration 66: epsilon_t = 0.4236, alpha_t = 0.1539, Z_t = 0.9883
Iteration 67: epsilon_t = 0.4444, alpha_t = 0.1117, Z_t = 0.9938
Iteration 68: epsilon_t = 0.4436, alpha_t = 0.1133, Z_t = 0.9936
Iteration 69: epsilon_t = 0.4437, alpha_t = 0.1131, Z_t = 0.9936
Iteration 70: epsilon_t = 0.4518, alpha_t = 0.0968, Z_t = 0.9953
Iteration 71: epsilon_t = 0.4254, alpha_t = 0.1504, Z_t = 0.9888
Iteration 72: epsilon_t = 0.4519, alpha_t = 0.0966, Z_t = 0.9954
Iteration 73: epsilon_t = 0.4620, alpha_t = 0.0761, Z_t = 0.9971
Iteration 74: epsilon_t = 0.4451, alpha_t = 0.1101, Z_t = 0.9940
Iteration 75: epsilon_t = 0.4165, alpha_t = 0.1686, Z_t = 0.9860
Iteration 76: epsilon_t = 0.4387, alpha_t = 0.1232, Z_t = 0.9925
Iteration 77: epsilon_t = 0.4424, alpha_t = 0.1158, Z_t = 0.9933
Iteration 78: epsilon_t = 0.4410, alpha_t = 0.1185, Z_t = 0.9930
Iteration 79: epsilon_t = 0.4593, alpha_t = 0.0816, Z_t = 0.9967
Iteration 80: epsilon_t = 0.4287, alpha_t = 0.1435, Z_t = 0.9898
Iteration 81: epsilon_t = 0.4384, alpha_t = 0.1238, Z_t = 0.9924
Iteration 82: epsilon_t = 0.4551, alpha_t = 0.0899, Z_t = 0.9960
Iteration 83: epsilon_t = 0.4400, alpha_t = 0.1206, Z_t = 0.9928
Iteration 84: epsilon_t = 0.4337, alpha_t = 0.1334, Z_t = 0.9912
Iteration 85: epsilon_t = 0.4024, alpha_t = 0.1978, Z_t = 0.9808
Iteration 86: epsilon_t = 0.4304, alpha_t = 0.1400, Z_t = 0.9903
Iteration 87: epsilon_t = 0.4249, alpha_t = 0.1514, Z_t = 0.9886
Iteration 88: epsilon_t = 0.4047, alpha_t = 0.1930, Z_t = 0.9817
Iteration 89: epsilon_t = 0.4139, alpha_t = 0.1739, Z_t = 0.9851
Iteration 90: epsilon_t = 0.4257, alpha_t = 0.1497, Z_t = 0.9889
Iteration 91: epsilon_t = 0.4369, alpha_t = 0.1270, Z_t = 0.9920
Iteration 92: epsilon_t = 0.4195, alpha_t = 0.1624, Z_t = 0.9870
Iteration 93: epsilon_t = 0.4092, alpha_t = 0.1836, Z_t = 0.9834
Iteration 94: epsilon_t = 0.3909, alpha_t = 0.2217, Z_t = 0.9759
Iteration 95: epsilon_t = 0.4227, alpha_t = 0.1558, Z_t = 0.9880
Iteration 96: epsilon_t = 0.4257, alpha_t = 0.1497, Z_t = 0.9889
Iteration 97: epsilon_t = 0.4297, alpha_t = 0.1416, Z_t = 0.9901
Iteration 98: epsilon_t = 0.4221, alpha_t = 0.1571, Z_t = 0.9878
Iteration 99: epsilon_t = 0.4179, alpha_t = 0.1657, Z_t = 0.9864
Iteration 100: epsilon_t = 0.4343, alpha_t = 0.1321, Z_t = 0.9913
Accuracy sur les données train: 100.0
Accuracy sur les données test: 76.33333333333333

```

```

In [20]: Z = np.cumprod(Z_values)

plt.figure(figsize=(12, 8))

plt.subplot(2, 2, 1)
plt.plot(epsilons)
plt.title("Erreur  $\epsilon_t$  par itération")
plt.xlabel("Itération t")
plt.ylabel(" $\epsilon_t$ ")

plt.subplot(2, 2, 2)
plt.plot(alphas)
plt.title("Poids  $\alpha_t$  par itération")
plt.xlabel("Itération t")
plt.ylabel(" $\alpha_t$ ")

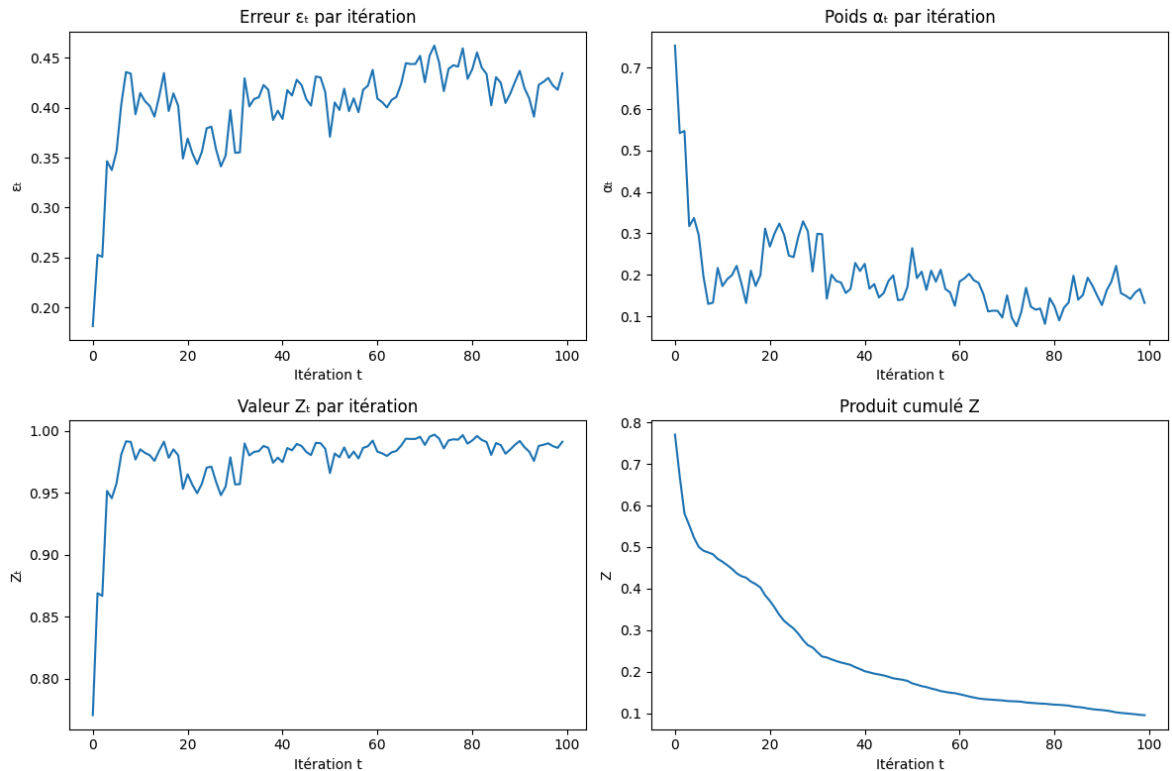
plt.subplot(2, 2, 3)

```

```
plt.plot(Z_values)
plt.title("Valeur  $Z_t$  par itération")
plt.xlabel("Itération  $t$ ")
plt.ylabel(" $Z_t$ ")

plt.subplot(2, 2, 4)
plt.plot(Z)
plt.title("Produit cumulé  $Z$ ")
plt.xlabel("Itération  $t$ ")
plt.ylabel(" $Z$ ")

plt.tight_layout()
plt.show()
```



Avec des arbres de décision de profondeur 1, on observe des résultats similaires entre les données d'entraînement et de test (78% et 79% d'accuracy). Cela montre que le modèle ne surapprend pas et qu'il reste assez stable au fil des itérations. À chaque itération, un nouvel arbre est ajouté au modèle, mais on remarque que le nombre d'itérations n'a pas un impact fort sur la précision tant que les classifieurs faibles restent simples. Cependant, lorsqu'on augmente la profondeur des arbres à 5, on observe du surapprentissage : l'accuracy atteint 99.85% sur le train, mais chute à 74.33% sur le test. Cela indique que les arbres profonds s'ajustent trop fortement aux données d'entraînement, ce qui empêche la généralisation.

En ce qui concerne l'erreur ϵ_t , on observe qu'elle tend à rester proche de 0.5 au fil des itérations. Cela signifie que les nouveaux classifieurs n'apportent plus d'amélioration significative, et que leurs performances sont proches du hasard (erreur de 50%). Pour les poids α_t , on voit une décroissance au cours des itérations : les premiers classifieurs ont des poids plus élevés car ils sont efficaces sur les données encore équilibrées uniformément. Plus on avance dans les itérations, plus les erreurs deviennent spécifiques, et les nouveaux arbres ont donc un poids plus faible car ils corrigent des cas

plus isolés.

Concernant le produit cumulé des normalisations Z , on observe aussi une diminution au fil des itérations. Cela signifie que le modèle global s'améliore : plus le modèle est bon, plus la somme pondérée des erreurs exponentielles diminue, ce qui entraîne une baisse de Z .

Conclusion :

Le bagging vise à réduire la variance du modèle en moyennant les prédictions de plusieurs arbres indépendants, tandis que le boosting cherche à réduire le biais en se concentrant sur les erreurs des itérations précédentes. Ainsi, le bagging est généralement plus robuste au surapprentissage, même avec des arbres profonds, grâce à l'effet de moyenne. Le boosting, lui, est plus sensible au surapprentissage si les arbres sont trop complexes ou trop nombreux, car il s'ajuste très finement aux données d'entraînement.